

GEMINI PDR

Graphene-Foil Electron-Emission Measurement Investigation for Novel Instrumentation

William Li

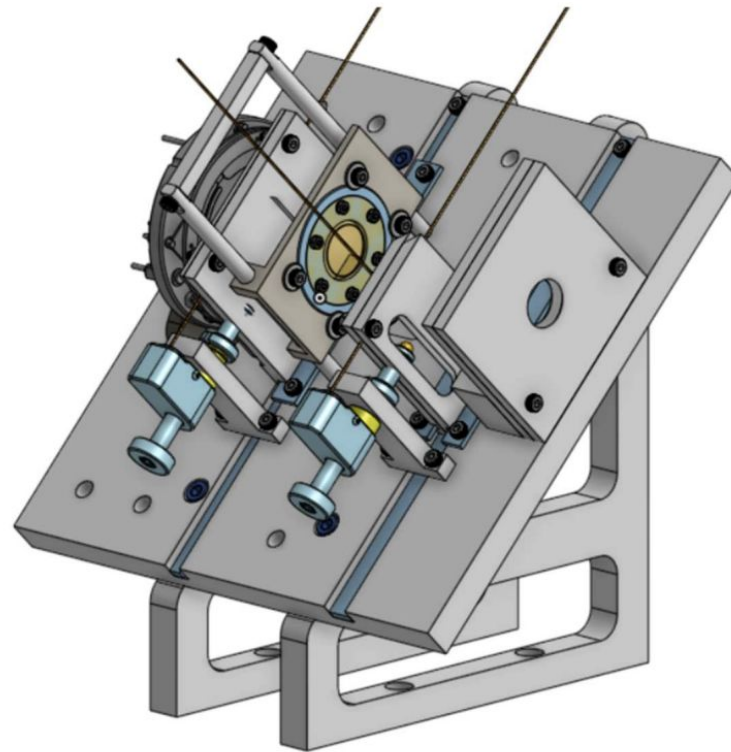
Junior Independent Work | Fall 2025

Motivation

Building off of the 2023-2024 AST250/251 lab class ultra-thin carbon foil analyzer.

Goals:

- Make the instrument more mechanically reliable
- Reduce mass and volume of the instrument
- Use the year to gather more data on foils



Credit: 2024 AST251 CDR
Ultra-thin Carbon Foil Analyzer - CDR Spring 2024.pdf

Science Motivation

The 2023-2024 lab class instrument specialized in:

- (1) Measuring foil thickness based on scattering angle
- (2) Observing wrinkles and pinhole defects.

Why do we care about foil thickness?

- Thinner foils have reduced angular scattering (Ebert et al. 2014)
- *Thinner foils seem to not reduce energy straggling, which reduces their ability to detect low energy particles (Allegrini et al. 2015, Allegrini et al. 2006)*

Could the “crud layer” be at play?

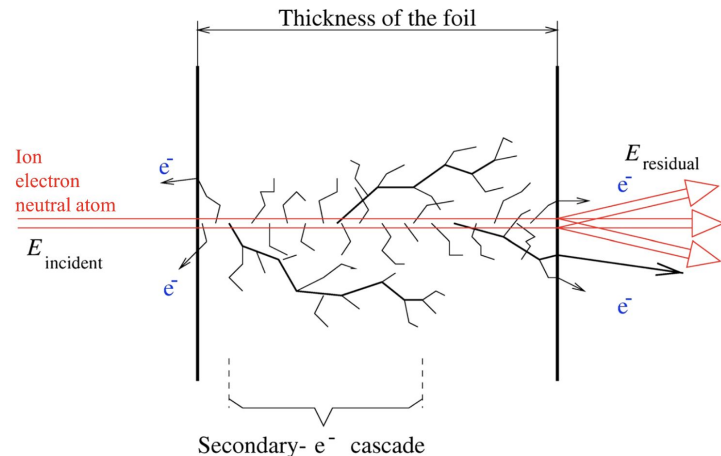


Figure 3. Schematic of the secondary electron emission process.

Allegrini et al. (2016)
Carbon foils for space plasma instrumentation

Important Science Parameters

- Secondary electrons are typically 2.1 eV (Allegrini et al. 2016)
- Scattering angles are typically $<12.5^\circ$
- Foil thickness measured in $\mu\text{g}/\text{cm}^2$
 - 3 foils for GEMINI: graphene, $0.6 \mu\text{g}/\text{cm}^2$, and $1.1 \mu\text{g}/\text{cm}^2$

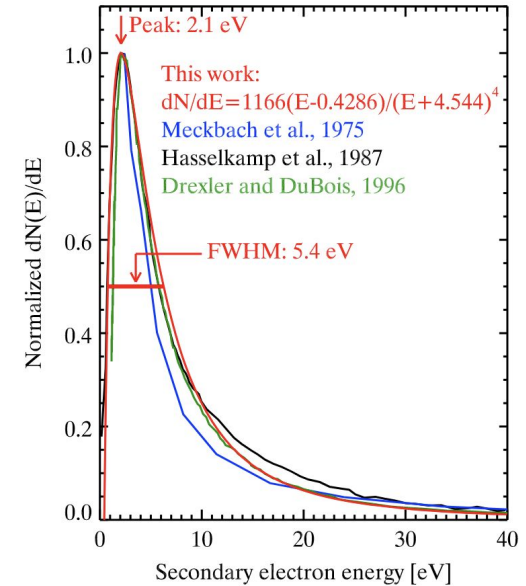
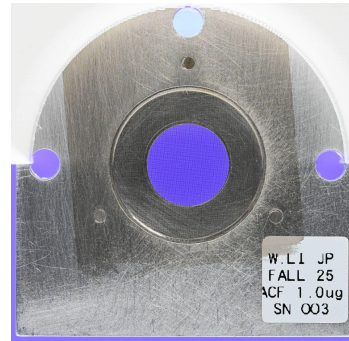
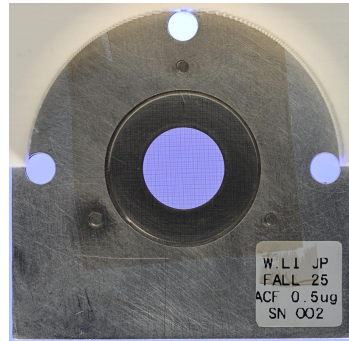
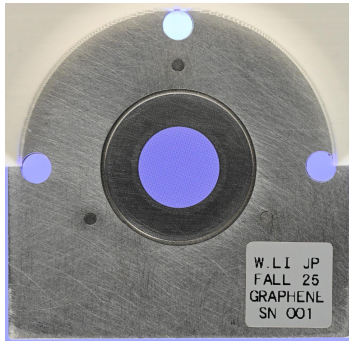
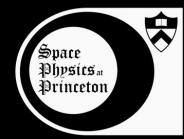


Figure 6. Secondary electron energy distribution from the proposed expression in red compared to measurements from Meckbach et al. [1975] in blue, Hasselkamp et al. [1987] in black, and Drexler and DuBois [1996] in green.

Allegrini et al. (2016)
Carbon foils for space plasma instrumentation



Project Goals

GEMINI will fulfill the lab class's proposed future science goals for graphene and carbon foils and measure:

- (1) Particle charge states after passing through the foil
- (2) Secondary electron emission rate and energy.

Then, GEMINI will connect these properties to (1) foil thickness and (2) scattering angle.

Next semester (or next year!) I want to compare the instrument's measured foil thickness with more direct measurement methods at the Imaging and Analysis Center and manipulate the foil on the atomic scale to try to improve on its properties.

Design

1. Deflected ions strike MCP

$$r > \alpha(\ell_1 + \ell_2 + \ell_3) + \frac{Vq\ell_2}{2Ed} (1 + \alpha^2) \left(\frac{\ell_2}{2} + \ell_3 \right)$$

2. Deflected ions miss DPs

$$\frac{d}{2} > \alpha(\ell_1 + \ell_2) + \frac{Vq\ell_2^2}{4Ed} (1 + \alpha^2)$$

3. Charge state pops. separated

$$\frac{V|q|\ell_2}{2Ed} (1 + \alpha^2) \left(\frac{\ell_2}{2} + \ell_3 \right) > 2\alpha(\ell_1 + \ell_2 + \ell_3)$$

4. Majority of angular scattering distribution strikes MCP

$$E > \frac{12.5t}{\tan^{-1} \left(\sqrt{\frac{\ln 2}{2}} \frac{r}{(s + \ell_1 + \ell_2 + \ell_3)} \right)}$$

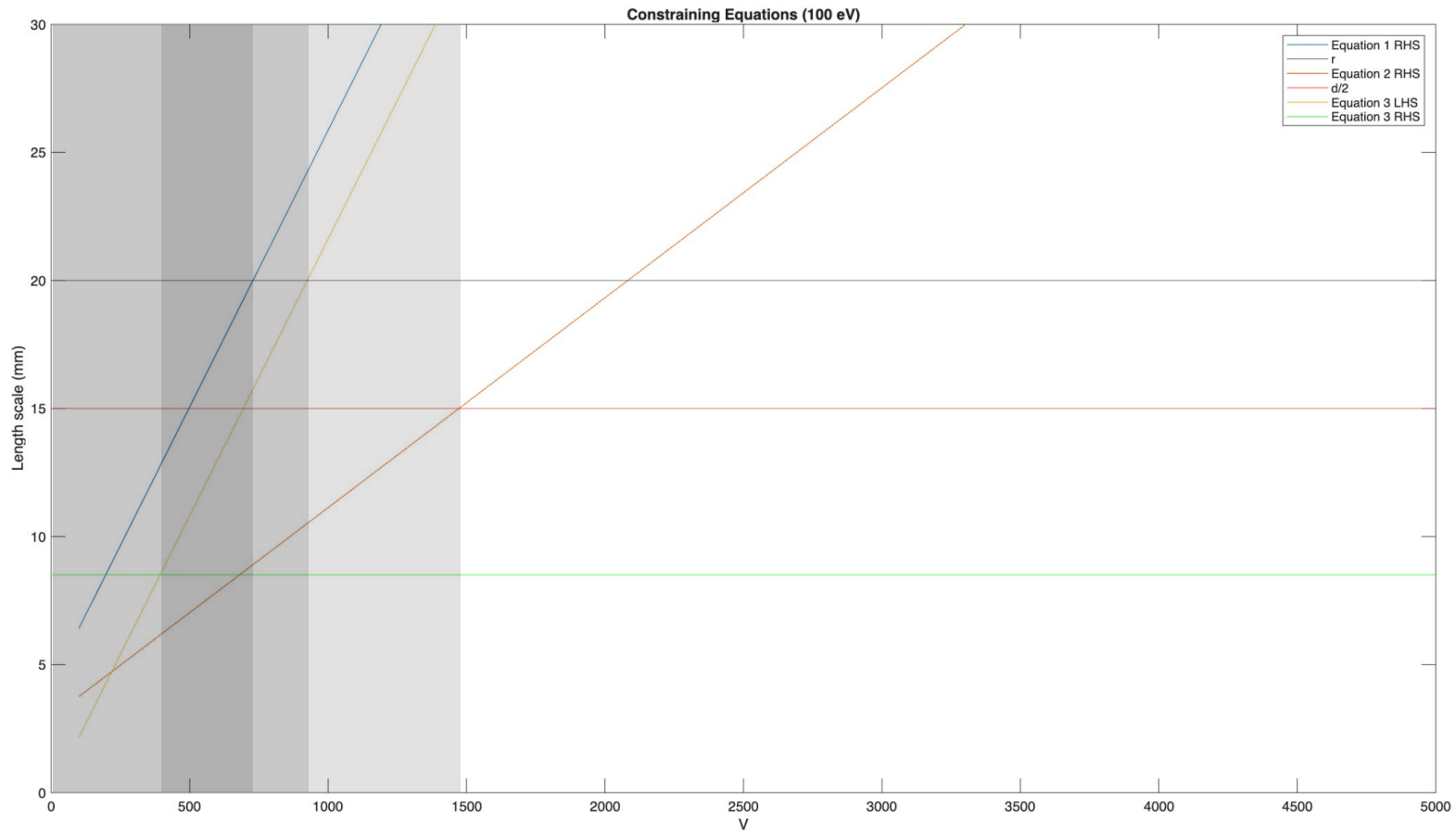
Constraining Equations

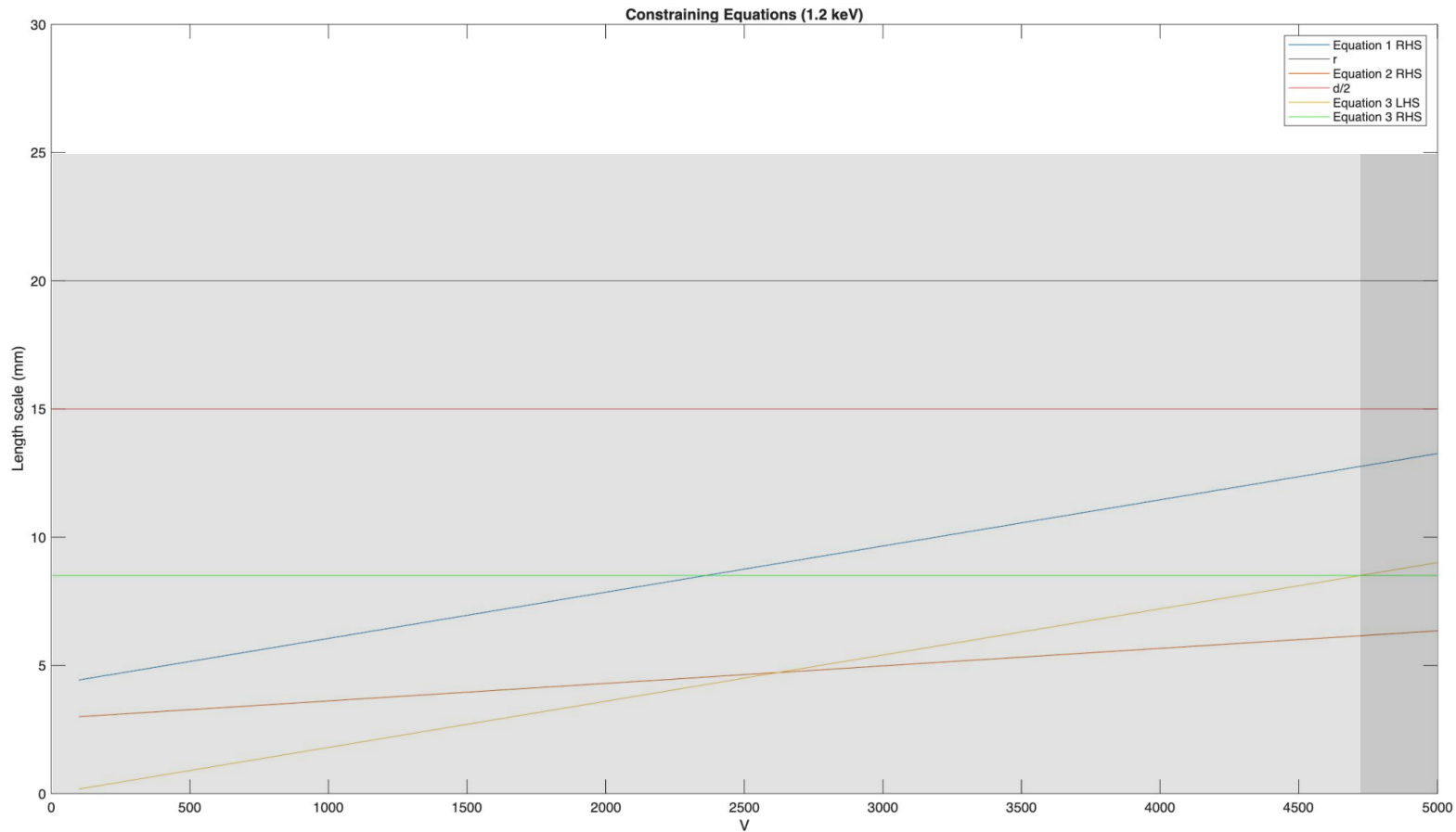
Credit: 2024 AST251 PDR
Ultra-thin Carbon Foils PDR - Fall 2023.pdf

Additional Constraints

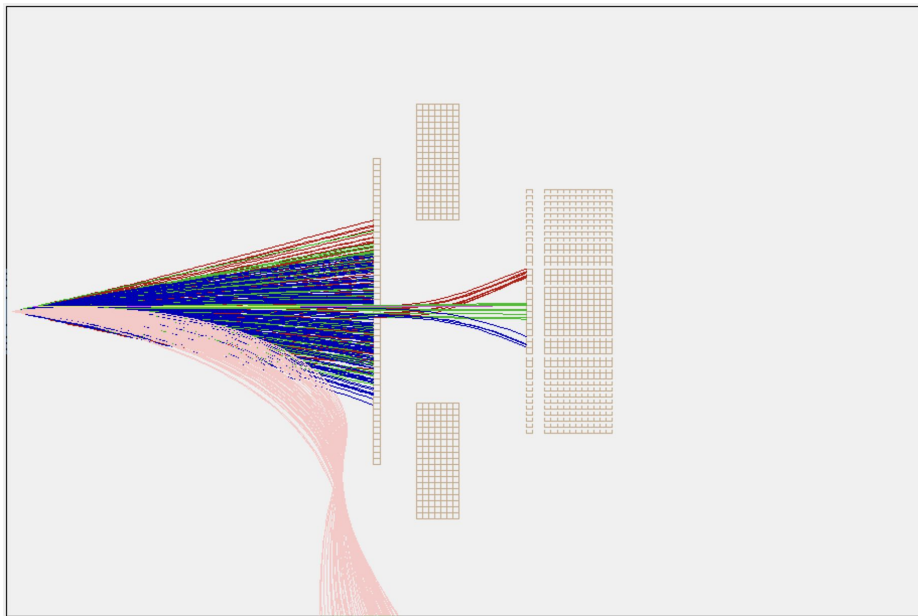
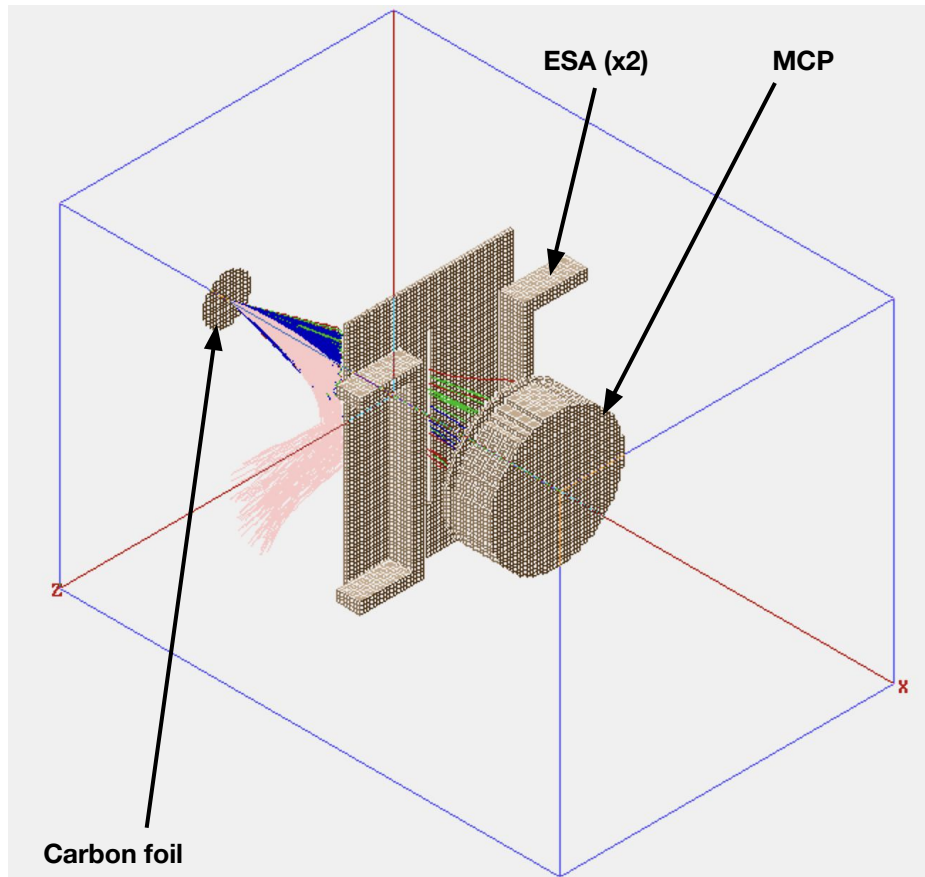
- $E_{\min} = 100 \text{ eV}$
- $V_{\max} = 5 \text{ kV}$
- $L_{\max} = 19/\tan(12.6^\circ)$
 - Constrains vertical scattering to MCP face
- $(5\text{kV ESA}) * (40 \text{ thou/kV}) = 200 \text{ thou}$
 - +10% margin = 220 thou = 5.588 mm

Parameter	Value
E (beam energy)	100 eV – 1.2 keV
V (ESA voltage)	<5000 V
r (radius of MCP)	20 mm
s (distance from carbon foil to slit plate)	67 mm
d (distance between ESA plates)	30 mm
α (w/t)	0.25
l_1 (distance from slit plate to ESA)	5.588 mm
l_2 (length of ESA plates)	6.825 mm
l_3 (distance from ESA plates to MCP)	5.588 mm





SIMION

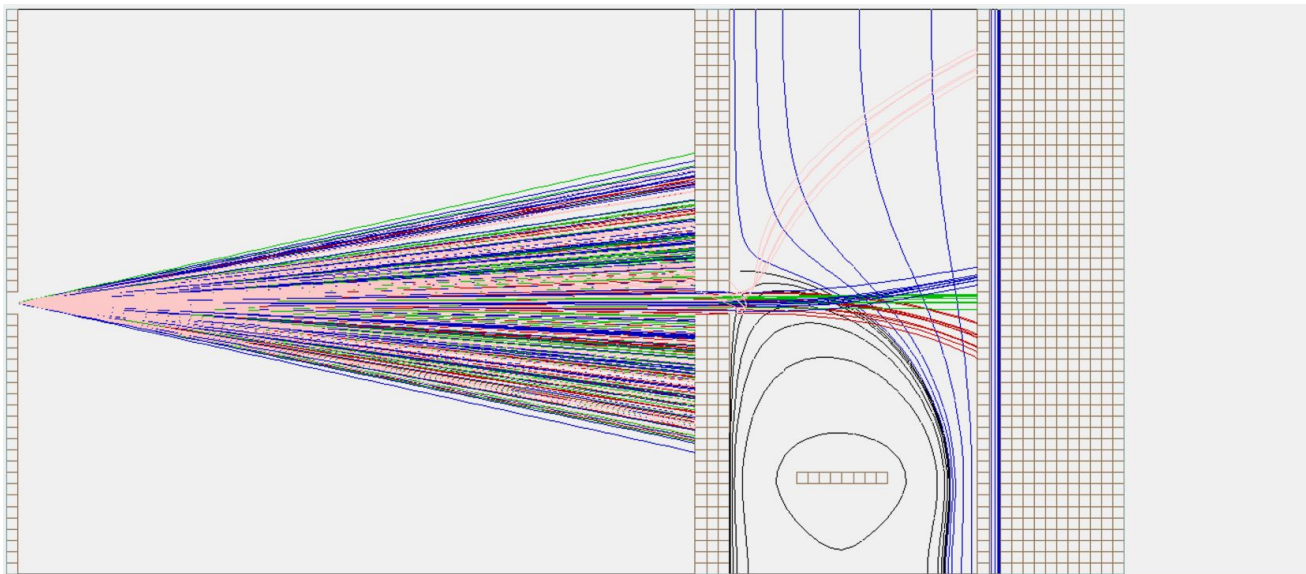


TOP VIEW

Charge State Separation on MCP
ESA = +250V

Secondary electrons (2eV) are very susceptible to E-fields!

SIMION - Secondary Electrons



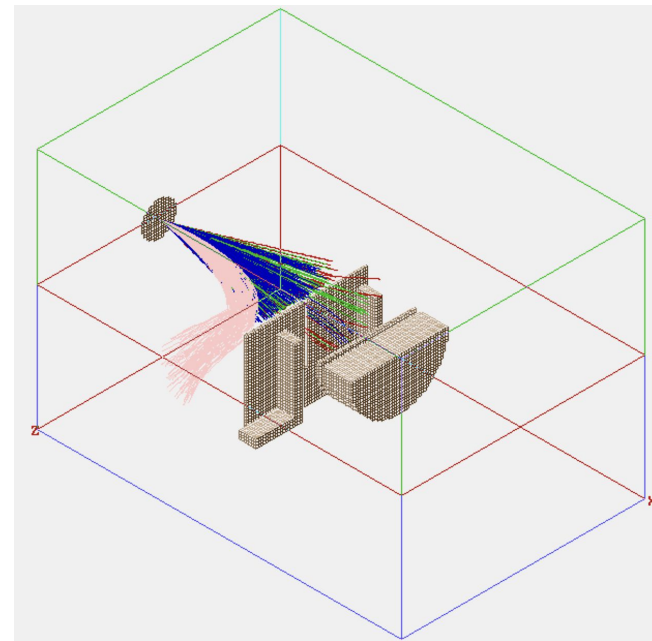
TOP VIEW

With one ESA at a lower voltage & a grid in front of the MCP at a small positive voltage, secondary electrons can be attracted to the MCP.

Science Observables

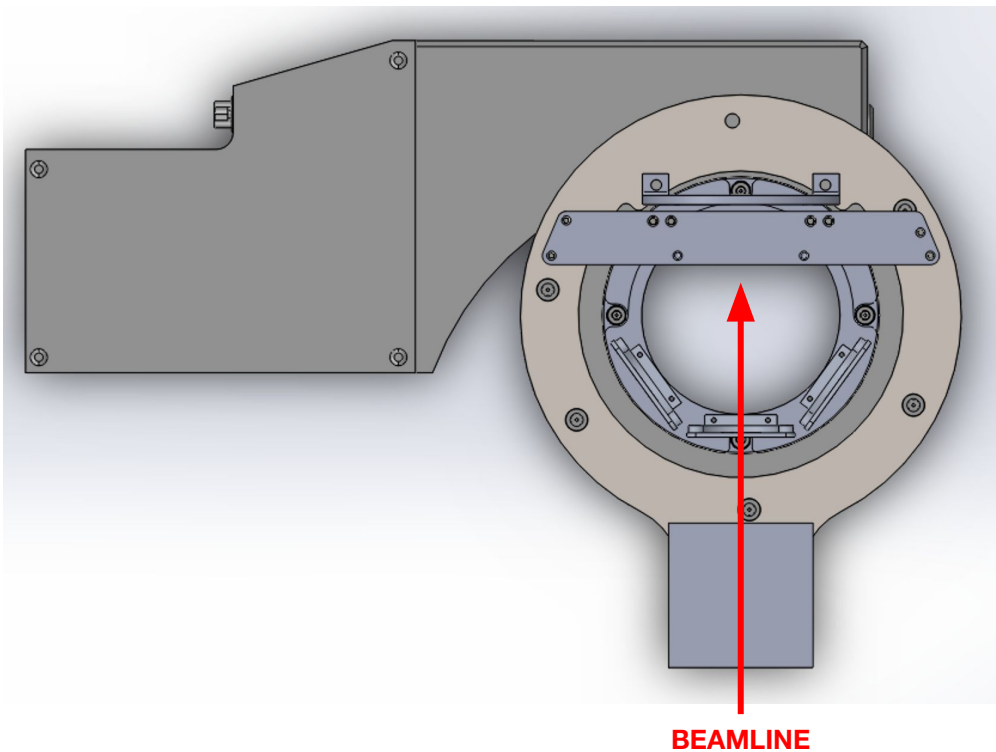
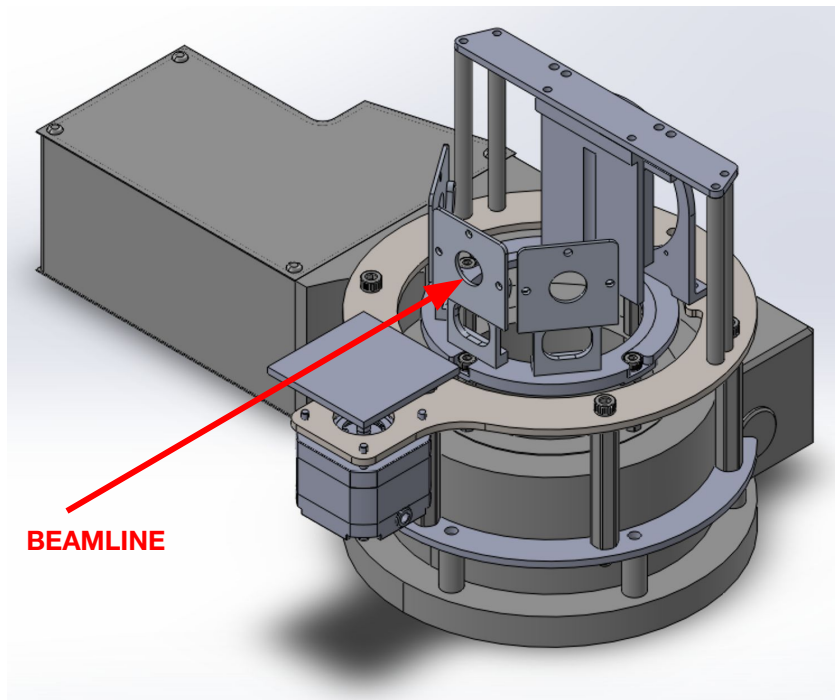
- Proportions of charge state of ions passing through carbon foil (horizontal distribution on MCP)
- Angular scattering (vertical distribution on MCP)
 - Can infer thickness from angular scattering!
- Secondary electron rate
- Secondary electron energy (based on location on MCP and voltage of grid)

All with one instrument and essentially no moving parts!

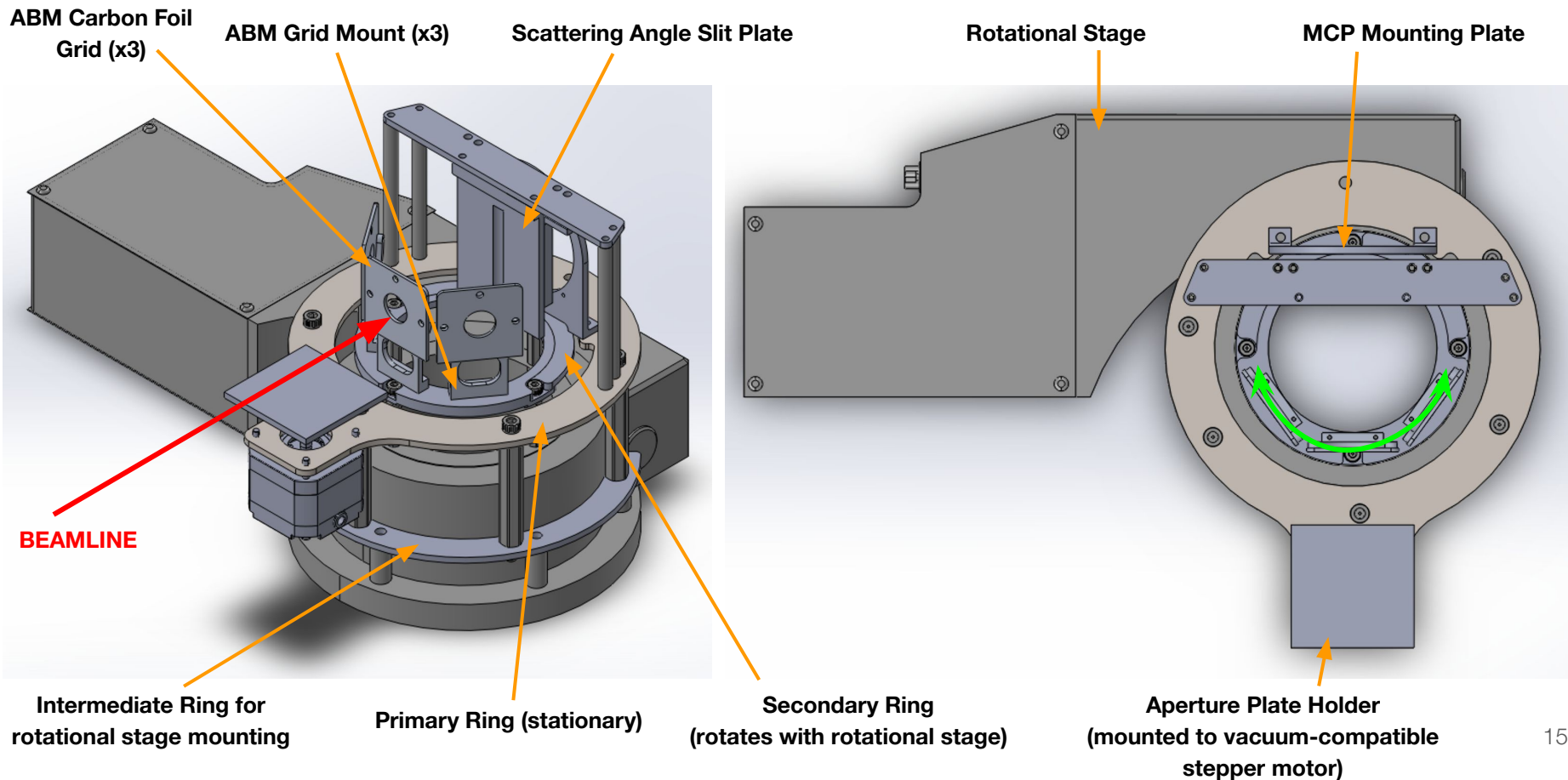


CUTAWAY VIEW

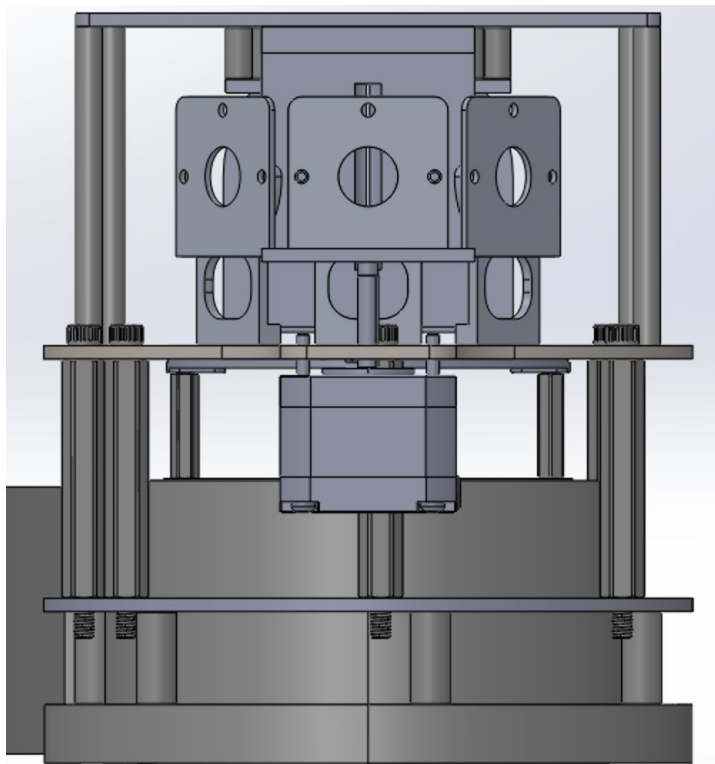
Instrument



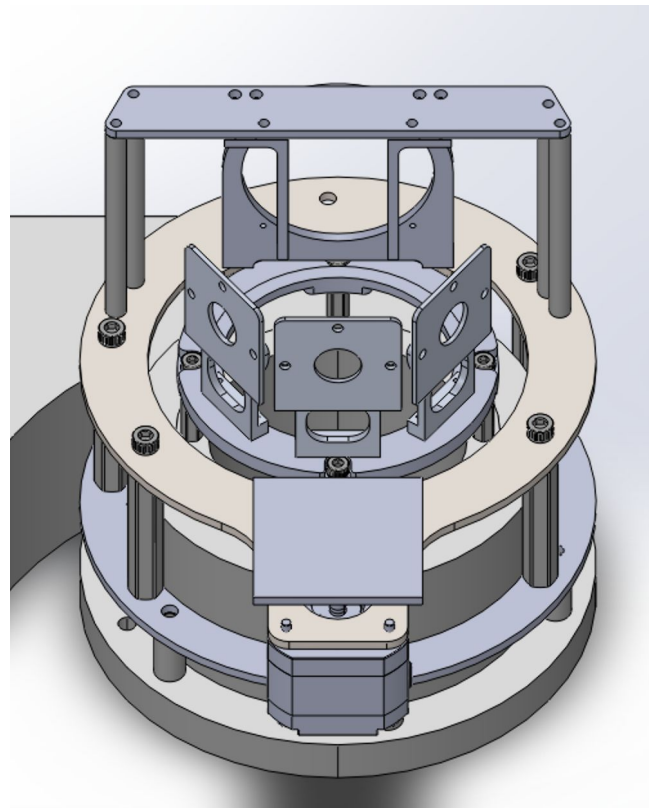
Instrument



Instrument



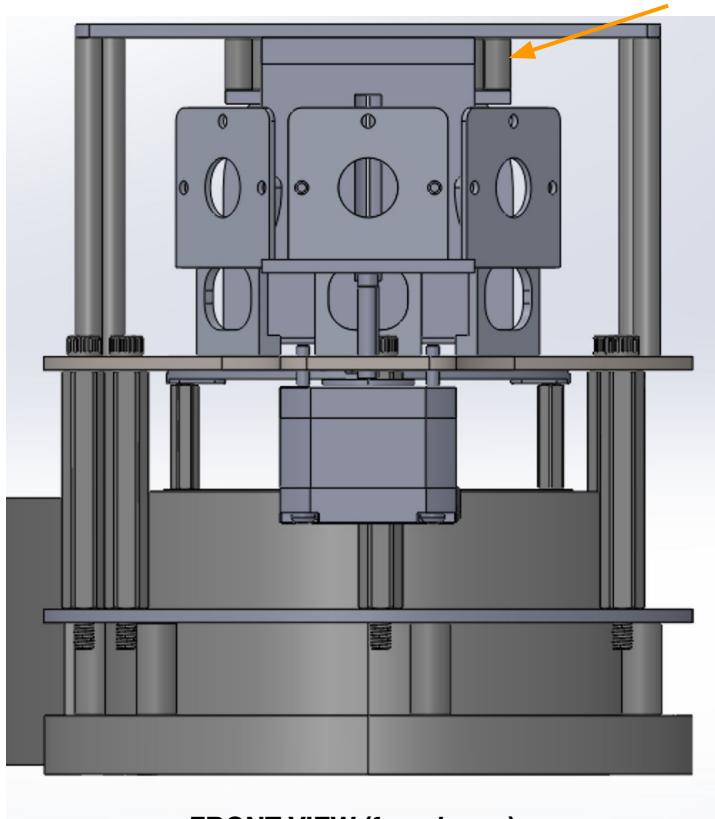
FRONT VIEW (from beam)



~FRONT VIEW (slit plate removed to see ESA plates)

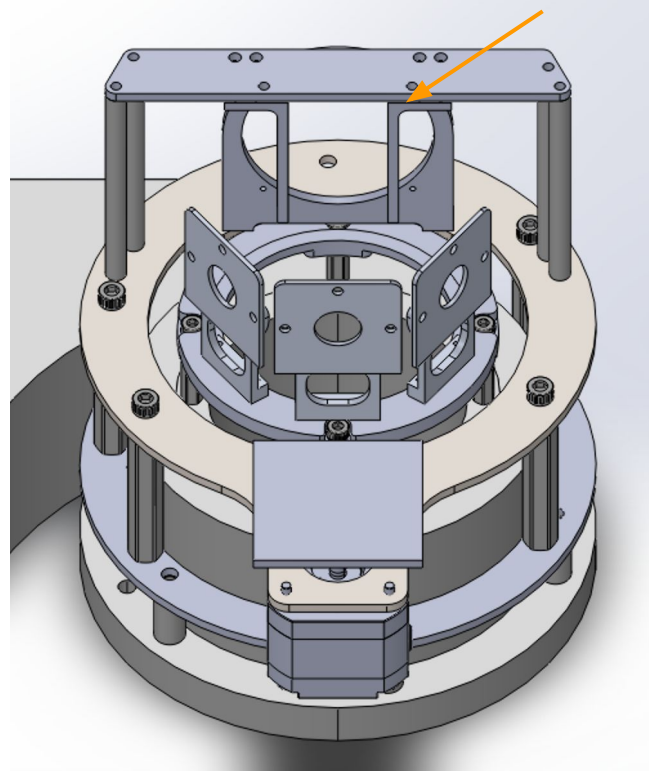
Instrument

Ceramic Standoffs for ESA



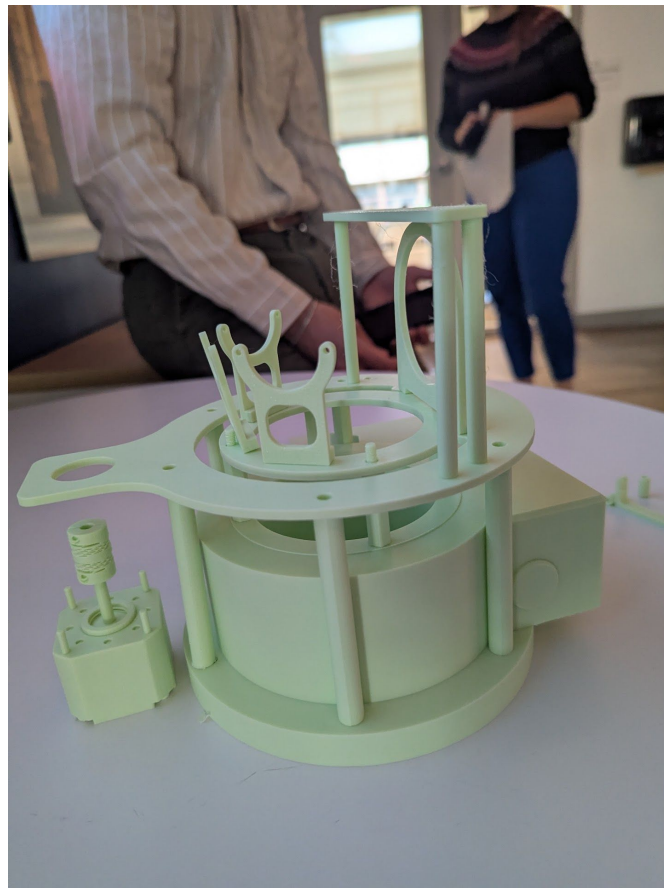
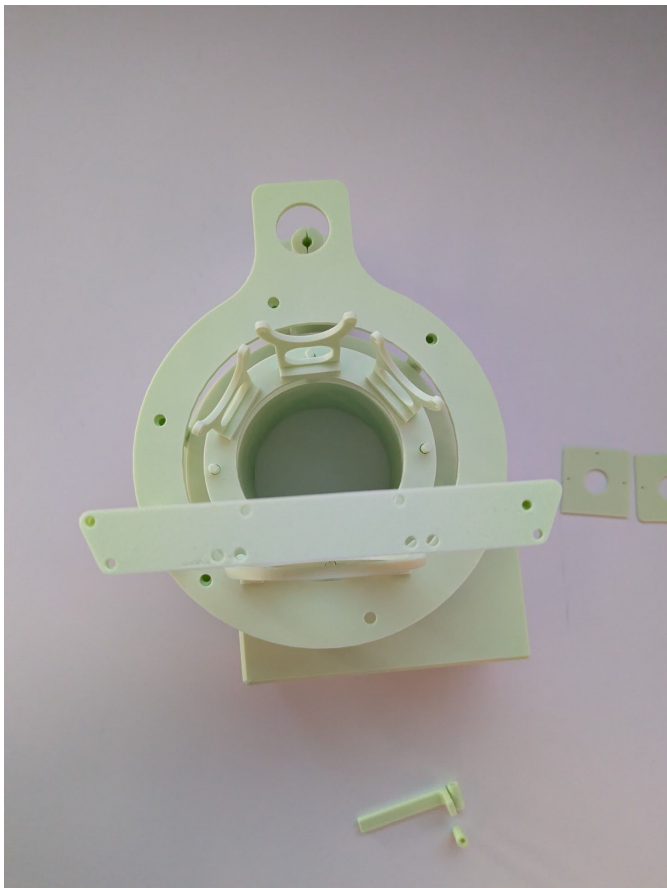
FRONT VIEW (from beam)

ESA Deflector Plate (x2)



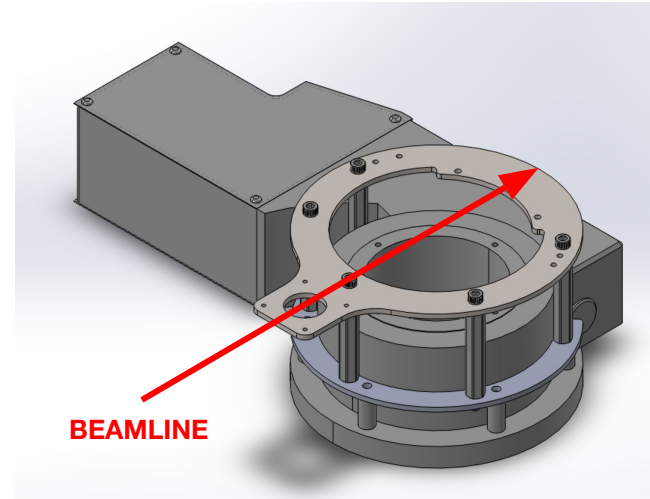
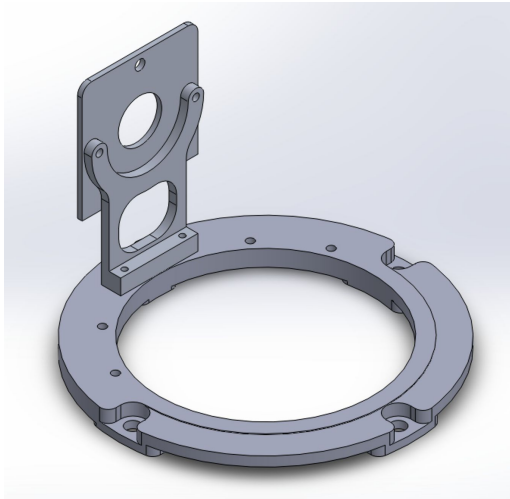
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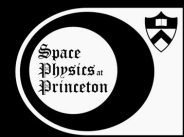
Test Print V1



Modifications from Test Print V1

- Thickened the rotational plate and counterbore bolts to prevent bolt heads from hitting ABM grid holder mounts or MCP mount
- Added an intermediate ring because the screws holding the rotational stage down must be vented
- Moved COM of ABM grid holder above the screw positions





Upcoming Steps

- **Week of 12/8**
 - Design a mounting mechanism for a grid in front of the MCP
 - Design aperture plate & shield in front of entire instrument to block the rest of the beam
 - Add tapped hole to ESA for supplying power
 - Purchase parts from McMaster (really frustrating, don't know how to access my awarded \$500)
 - Make CNC files for machining
- **Week of 12/15**
 - Machine parts and clean to prepare for bakeout

Conclusion